

Deadlock detection in Distributed systems:

System Models

CST402 Distributed Computing

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Deadlock:

A deadlock can be defined as a condition where a set of processes request resources that are held by other processes in the set.

System Model:

Overview:

- Consists of a set of process connected via a communication network.
- Communication delay is finite but unpredictable.

Components of a Distributed Program:

- Consists of n asynchronous processes denoted as $P_1, P_2, \dots, P_i, \dots, P_n$.
- Communicate via message passing.
- Assumes each process operates on a different processor.

System Characteristics:

- No common global memory among processors.
- Communication occurs only through message passing.
- Absence of a physical global clock for instantaneous access for processes.

Components of a Distributed Program:

- Consists of n asynchronous processes denoted as $P_1, P_2, \dots, P_i, \dots, P_n$.
- Communicate via message passing.
- Assumes each process operates on a different processor.

Communication:

- May deliver messages out of order, lost, garbled or duplicated.
- Happens due to timeout and retransmission.
- Processor and communication link failure can lead to disruptions.

Representation:

- Modeled as directed graph.
- Vertices -> Processes
- Edges -> Unidirectional communication channels

Assumptions:

- System has only reusable resources.
- Processes require exclusive access to resources, meaning only one process can utilize a resource at any given time.
- Each resource exists as a single instance within the system.

Process States:

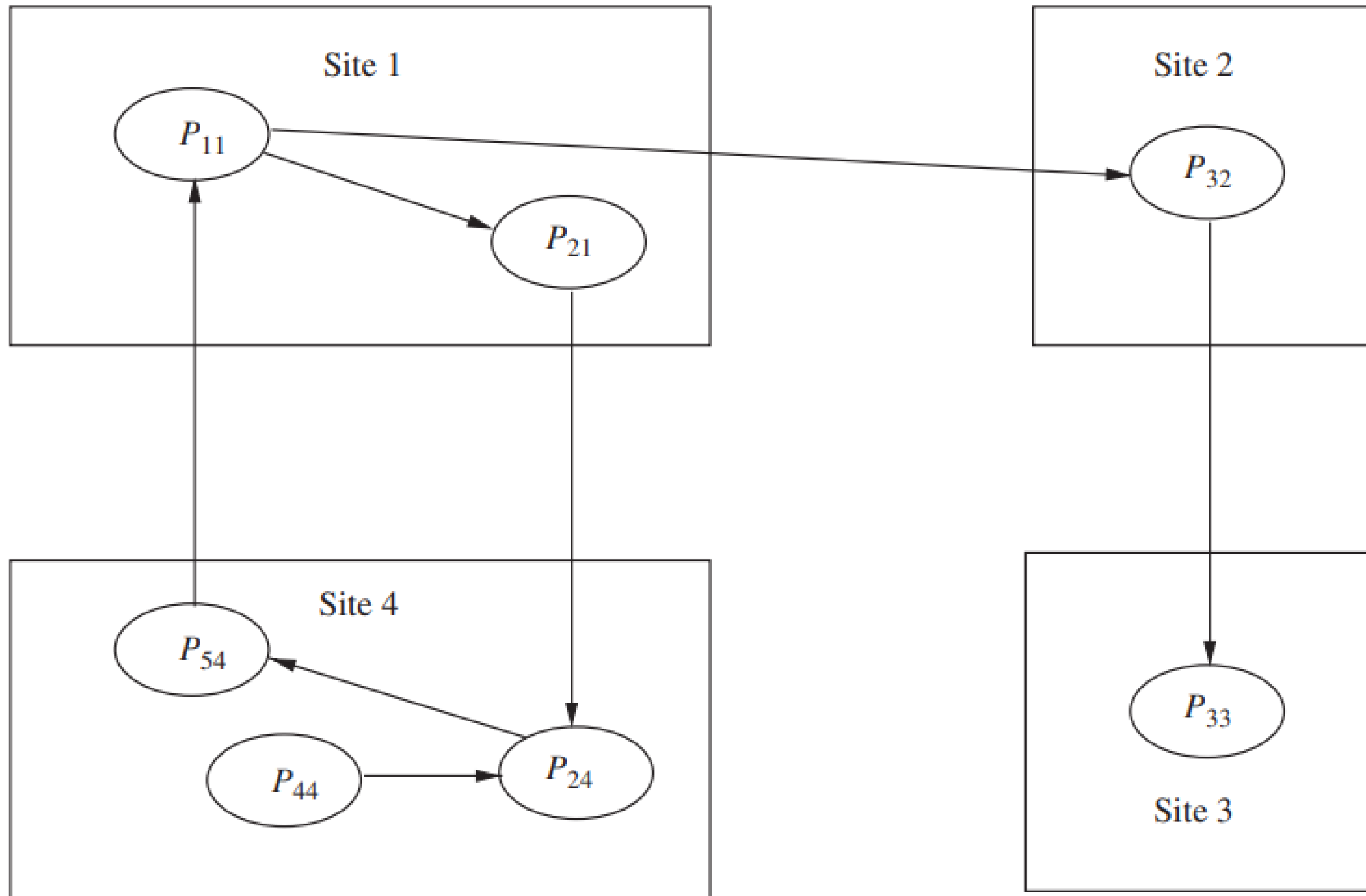
- Can exist in one of two states: running or blocked.
 - ***running state***(*active* state):
 - Process possesses all needed resources.
 - Actively executing or ready for execution.
 - ***blocked state***(*active* state):
 - Process lacks needed resources.

Wait-for graph(WFG)

- In distributed systems, the ***system's state*** can be represented by a directed graph known as a ***wait-for graph (WFG)***.
- Nodes in the WFG correspond to processes within the system.
- Directed edges indicate a blocking relationship between processes: if process P1 is blocked and waiting for process P2 to release a resource, there exists a directed edge from P1 to P2 in the WFG.

Deadlock detection:

- Deadlock in a distributed system can be identified by examining the WFG.
- A system is considered deadlocked if and only if there exists a directed cycle or knot in the WFG.



Example of WFG